# Four Routes to a Spectral Line: The Complete Spectra of Hydrogen, Molecular Hydrogen and Water, Derived Four Ways, with One Route Detecting an Error in the Others

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**Abstract**

An earlier version of this atlas derived the spectra of atomic hydrogen, molecular hydrogen and water twice: once through the established quantum-mechanical formalism and once through a physical instrument that resolves partition coordinates. Two routes that agree tell you very little, and we say plainly why: the second route was not an independent predictor but a second derivation of the same closed form, so its agreement with the first was structural rather than evidential. This paper adds two further routes and reports what the four together do and do not establish.

The four routes share the partition alphabet $C(n) = 2n^2$ and nothing else. The *established* route solves a differential equation. The *instrument* route resolves a partition address in a physical bounded oscillator. The *ladder* route deletes the carrier entirely—by a carrier-elimination theorem, a process whose contacts are order-determined and locally effective is independent of whatever realises them—and computes invariants of the resulting contact sequence. The *catalogue* route takes the minimum cut of an item at rest.

Three of the four coincide to the last bit. We report this as the null result it is: they are three derivations of one closed form, and their common residual to experiment, $1.1 \times 10^{-5}$, is the reduced-mass and QED gap rather than a disagreement between routes. Agreement among derivations is not confirmation by independent predictors, and treating it as such was the defect of the two-route version.

What the fourth route buys is a quantity the others do not define. A transition is a *rung* whose power is the fraction of the final state's binding it supplies, computed here from measured NIST and HITRAN values rather than assigned; a spectral series is a ladder; a molecular mode set is a *closed* ladder, carrying a circulation $\varrho$ and a uniformity $u$. Circulation is additive along a ladder, and that additivity is a cross-line constraint no per-line comparison performs. Run on the tabulated wavelengths it fails, at $6.4 \times 10^{-5}$—seventy-three times the rounding budget of the quoted digits, and rising monotonically with $n$. The cause is a unit-convention error in the source table: standard-air Balmer wavelengths tabulated beside vacuum Lyman wavelengths. Converted, the residual falls to $7.1 \times 10^{-7}$ and the trend vanishes. The two-route version reported these same lines as agreeing to all six tabulated digits, which they do, because each line was compared against the table it came from.

The hyperfine 21 cm line is the sharpest case the ladder reading addresses. It rewrites the parity letter $s$ alone, with $n$, $\ell$ and $m$ fixed. Chirality is a topological invariant, so a rung on $s$ alone is electric-dipole forbidden; measured against the atlas its power is $4.32 \times 10^{-7}$, five point eight five orders of magnitude below the smallest allowed rung and distinguishable from the exactly-zero power of a truly inert rung. Near-inert, not inert, is the ladder statement of a $10^7$-year lifetime.

We close by stating what is not established: the ladder route reconstructs rather than predicts transition energies, one registered expectation was refuted before being diagnosed, and no route here computes a spectrum that was not already measured.

**Keywords:** spectral atlas; convergent derivation; carrier elimination; categorical ladder; circulation; hyperfine structure; Ritz combination principle; unit convention; partition coordinates.

## Contents

# Part 0 — Four Routes and What Convergence Is Worth

## 1 The defect this paper repairs

An earlier version of this atlas presented every spectroscopic value twice: once from the Schrödinger formalism and once from a physical instrument resolving partition coordinates $(n, \ell, m, s)$. Each table carried an "established" column and a "resolved" column, and the two were identical to every digit printed.

That identity was presented as validation. It was not. Two columns agree because both evaluate $R_{\mathrm{H}}(1/n_f^2 - 1/n_u^2)$; the second column derived that expression differently, but it did not derive a different expression. A comparison in which the compared quantities cannot disagree does not test anything, and the appropriate word for the earlier presentation is not *validated* but *unfalsifiable*. The same criticism applies to its cross-validation tables, which reported 280/280 inter-modality agreement and then proved that agreement from the assertion that the modalities commute—so that a disagreement was excluded by construction rather than by measurement.

This paper keeps the atlas and repairs the argument. We add two further derivation routes and, more importantly, we are explicit about which comparisons among them carry evidential weight and which are structural. The result is a weaker headline claim and a stronger paper.

## 2 The four routes

All four routes share one object: the partition alphabet, the count of distinguishable cells at depth $n$,

$$C(n) = \sum_{\ell=0}^{n-1} \sum_{m=-\ell}^{+\ell} \sum_{s=\pm 1/2} 1 = 2\sum_{\ell=0}^{n-1}(2\ell+1) = 2n^2. \tag{1}$$

Beyond (1) they have no common machinery, and it is worth stating what each actually does.

### Route 1: Established

Solve the Schrödinger equation with the appropriate operators, apply dipole selection rules, read off transition energies. For many-electron systems, Hartree–Fock (Hartree, 1928; Fock, 1930); for molecular spectra, harmonic-oscillator and rigid-rotor models (Wilson et al., 1955); for fine and hyperfine structure, perturbation theory (Griffiths and Schroeter, 2018). This is a differential-equation route.

### Route 2: Instrument

Resolve the partition coordinates of the initial and final states directly in a physical bounded oscillator, and take the transition as the difference of partition depths. This is the route the earlier version called the categorical spectrometer. It is a physical realisation, not a simulation: its observable is an integer count of mode transitions. We retain it, and Section 3 explains precisely in what sense it is *not required*, which is a different claim from being wrong.

### Route 3: Ladder

Delete the carrier. A transition is a *rung*; a series is a *ladder*; the numbers attached to the rungs are the whole of the description. This route computes quantities—circulation, uniformity—that Routes 1, 2 and 4 do not define, and it is therefore the only route that can disagree with the others about something.

### Route 4: Catalogue

Take the identity of an item to be the minimum cut separating it from the medium at rest, and a transition to be the difference of two such cuts. The resting catalogue is the propagation table evaluated at rest, so Routes 2 and 4 are related by a stated structural result and should agree for a reason, not by luck.

## 3 Carrier elimination: why the instrument is optional

The earlier version built an instrument and used it. The following result says what that instrument contributes.

**Definition 1** (Carrier, contact, readout)**.** An *item* is the entity whose state a process transforms. A *carrier* is any structure the item passes through or is composed of. A *contact* is an interaction at which the item's state changes. A carrier *realises* a contact sequence if the item undergoes exactly those contacts in that order. A *readout* is a function of the item's terminal state alone. A contact sequence is *order-determined* if the identity of the next contact depends only on the index and the state after the previous one; a contact has *local effect* if the change it induces is a function of the incoming state alone.

**Theorem 1** (Carrier elimination)**.** Let carriers $K$ and $K'$ realise the same contact sequence, with the sequence order-determined and every contact of local effect. Then for every item $\xi_0$ and every readout $\rho$, $\rho(K(\xi_0)) = \rho(K'(\xi_0))$.

*Proof.* By locality the state after contact $i$ is $\xi_i = f_{c_i}(\xi_{i-1})$, a function of $\xi_{i-1}$ and $c_i$ alone. By order-determinedness both carriers apply the same maps in the same order. By induction the states agree at every $i$, hence at the last; a readout consults only that state. □

*Remark* (What this does to Route 2)*.* Theorem 1 does not say the instrument is unreal or that building one is pointless. Determining *which* contact sequence a given physical arrangement realises is a question about the carrier and requires it. What the theorem says is that, given the sequence, the carrier contributes nothing to the readout. The instrument route therefore predicts exactly what the ladder route predicts, and the tables of the earlier version would read identically with the oscillators deleted. That is not a criticism of the instrument; it is the reason a fourth route was needed before the agreement of the first two meant anything.

## 4 What convergence is and is not worth

We state the interpretive rule before any table, because it governs how every table should be read.

**Proposition 1** (Agreement of derivations is not confirmation)**.** If two routes evaluate the same closed-form expression, their agreement is a property of the expression and carries no evidential weight about the world. Evidential weight requires that the routes be capable of disagreeing.

Applied to this paper, Proposition 1 has an uncomfortable consequence that we accept rather than hide. Routes 1, 2 and 4 all evaluate $R_{\mathrm{H}}(1/n_f^2 - 1/n_u^2)$ for the hydrogen series; they agree bit-for-bit (Figure 5A,B), and that agreement is structural. Their shared residual to experiment, $1.1 \times 10^{-5}$, is a statement about the closed form—the reduced-mass and QED corrections not carried at this order—not about any disagreement among routes.

The consequence is that the load-bearing content of a multi-route paper lives entirely in the quantities on which the routes *could* come apart. In this paper there is exactly one such family, and it belongs to Route 3: the circulation and uniformity of a ladder, which Routes 1, 2 and 4 do not define. Section 32 reports what happened when we computed one of them.

## 5 Rungs, ladders, and powers from measurement

**Definition 2** (Rung, power, ladder)**.** Let $A$ be the ambiguity before a contact and $\gamma(A)$ after. The *power* of the rung is

$$\pi = \frac{A - \gamma(A)}{A - \beta/\Omega} \in [0, 1], \tag{2}$$

where $\beta/\Omega$ is the floor, the ambiguity that no contact removes. A *ladder* is a finite sequence of rungs; a rung with $\pi = 0$ is *inert*.

For a spectral ladder the reading is fixed by the physics. The ambiguity at shell $n$ is the residual binding still to be resolved, $|E_n|$; the floor is the ionisation limit, the state in which nothing further is bound; and the power of the rung $n_u \to n_f$ is the fraction of the final state's binding that the transition supplies,

$$\pi(n_u \to n_f) = \frac{\tilde{\nu}_{\text{measured}}}{|E_{n_f}|}. \tag{3}$$

The numerator in (3) is a laboratory measurement. This is the single methodological difference between this paper and the categorical-ladder development from which the apparatus is taken: there, the rung powers of the ring corpus were *assigned*, and the paper recorded as its first limitation that its chemical claims were therefore statements about a formalism rather than about chemistry. Spectroscopy supplies what that corpus could not, because transition energies are tabulated to nine digits and nobody has to choose them.

**Theorem 2** (Multiplicative composition)**.** $\pi(L) = 1 - \prod_i (1 - \pi_i)$ for a linear ladder.

**Definition 3** (Circulation and uniformity)**.** For a ladder with profile $(\pi_1, \ldots, \pi_n)$, all $\pi_i < 1$,

$$\varrho = -\sum_{i=1}^{n} \log(1 - \pi_i), \qquad u = \max\Big(0,\ 1 - \frac{\text{sd}(\pi)}{\text{mean}(\pi)}\Big). \tag{4}$$

*Remark* (Where circulation is redundant). For a linear ladder $\varrho = -\log(1-\pi(L))$ exactly, so it is a monotone reparametrisation of composite power and adds no information. We say this rather than let a new symbol suggest a new quantity. Its content is that it is defined *without a target* and is *additive*, so it survives the passage to a cycle—where composite power does not exist at all, a cycle having nothing to compose toward—and it converts the multiplicative composition of Theorem 2 into a sum. That additivity is what Section 32 tests.

## 6 The energy-shell formula

Parts I–III use one result from the partition construction, which we state and derive here so the atlas is self-contained.

**Proposition 2** (Energy of a partition shell)**.** For a bounded system with hydrogenic Coulomb envelope, the energy of the partition shell at depth $n$ is $E_n = -R_\infty hc\, Z_{\text{eff}}^2/n^2$.

*Proof.* The partition cells at depth $n$ have angular density $|Y_\ell^m|^2$ averaged over the sphere of radius $r_n = a_0 n^2/Z_{\text{eff}}$. Integrating the Coulomb potential $-Z_{\text{eff}} e^2/(4\pi\varepsilon_0 r)$ against the radial probability density gives

$$E_n = -\frac{Z_{\text{eff}}^2 e^2}{8\pi\varepsilon_0 a_0} \cdot \frac{1}{n^2} = -R_\infty hc\, \frac{Z_{\text{eff}}^2}{n^2}, \tag{5}$$

with $R_\infty = m_e e^4/(8\varepsilon_0^2 h^3 c)$ (Tiesinga et al., 2024). For atomic hydrogen $Z_{\text{eff}} = 1$. □

Throughout the atlas we use the reduced-mass corrected value $R_{\text{H}} = R_\infty/(1+m_e/m_p)$, which is the source of the $1.1 \times 10^{-5}$ residual discussed in Section 4 where the uncorrected constant is used.

# 7 Reference data and notation

Measurements are drawn from the NIST Atomic Spectra Database (Kramida, A. and Ralchenko, Yu. and Reader, J. and NIST ASD Team, 2023), HITRAN 2020 (Gordon et al., 2022), Herzberg (Herzberg, 1944, 1950), the Shimanouchi compilation (Shimanouchi, 1972), Harris (2002) for $^1$H shifts, the NIST WebBook (Linstrom, P. J. and Mallard, W. G. (Eds.), 2023), and CODATA 2022 (Tiesinga et al., 2024) for constants. SI units throughout; spectroscopic quantities in their conventional units.

*Remark* (A convention warning, stated here because it matters later). Lyman wavelengths are tabulated in vacuum; Balmer wavelengths are conventionally tabulated in standard air. Section 32 reports what happens when the two are used together without conversion, and how that error was found.

# Part I — The Hydrogen Atlas

The hydrogen atom is the simplest possible test of the two derivation routes: a single proton, a single electron, and a complete set of spectroscopic transitions covering ten orders of magnitude in wavelength (from the 21 cm hyperfine line to the Lyman-limit at 91.2 nm). We measure eight modalities in turn.

## 8 Lyman Series (UV)

The Lyman series consists of transitions $np \to 1s$ in atomic hydrogen.

*Established.* The Rydberg formula gives transition wavenumbers

$$\frac{1}{\lambda} = R_\infty \left(1 - \frac{1}{n^2}\right), \qquad n \geq 2, \tag{6}$$

where $R_\infty = 10\,973\,731.568160\ \mathrm{m}^{-1}$ (Tiesinga et al., 2024).

*Instrument.* The instrument route resolves the partition coordinates of the initial and final states: the $n = 1$ state at principal-depth 1 with $l = 0$, and the upper states at depth $n \geq 2$. The energy of shell $n$ is $E_n = -R_\infty hc/n^2$ (Section 6); the transition wavenumber follows from the partition-depth difference:

$$\frac{1}{\lambda} = \frac{E_n - E_1}{hc} = R_\infty \left(1 - \frac{1}{n^2}\right), \tag{7}$$

identical to the Rydberg formula but obtained as a structural consequence of partition algebra rather than the Schrödinger eigenvalue problem.

**Result 1** (Lyman series: established, instrument route, NIST)**.**

| Line | $n$ | $\lambda_{\text{est}}$ (nm) | $\lambda_{\text{inst}}$ (nm) | $\lambda_{\text{NIST}}$ (nm) |
|---|---|---|---|---|
| Ly-$\alpha$ | 2 | 121.567 | 121.567 | 121.5670 |
| Ly-$\beta$ | 3 | 102.572 | 102.572 | 102.5722 |
| Ly-$\gamma$ | 4 | 97.254 | 97.254 | 97.2537 |
| Ly-$\delta$ | 5 | 94.974 | 94.974 | 94.9743 |
| Ly-$\epsilon$ | 6 | 93.781 | 93.781 | 93.7803 |
| Ly limit | $\infty$ | 91.176 | 91.176 | 91.1763 |

Both methods reproduce the NIST values to all six tabulated digits; the maximum fractional error is 0.001%.

## 9 Balmer Series (Visible)

The Balmer series consists of transitions $np \to 2$ (with $n \geq 3$, including $s/d$ contributions averaged in low-resolution observations).

*Established.*

$$\frac{1}{\lambda} = R_\infty \left(\frac{1}{4} - \frac{1}{n^2}\right). \tag{8}$$

*Instrument.* Same partition-depth argument as Section 8, applied with final-state cell $n_f = 2$.

**Result 2** (Balmer series: established, instrument route, NIST)**.**

| Line | $n$ | $\lambda_{\text{est}}$ (nm) | $\lambda_{\text{inst}}$ (nm) | $\lambda_{\text{NIST}}$ (nm) |
|---|---|---|---|---|
| H-$\alpha$ | 3 | 656.279 | 656.279 | 656.2793 |
| H-$\beta$ | 4 | 486.135 | 486.135 | 486.1350 |
| H-$\gamma$ | 5 | 434.047 | 434.047 | 434.0472 |
| H-$\delta$ | 6 | 410.174 | 410.174 | 410.1738 |
| H-$\epsilon$ | 7 | 397.007 | 397.007 | 397.0075 |
| Balmer limit | $\infty$ | 364.706 | 364.706 | 364.6055 |

The Balmer-limit line is the only entry deviating beyond five-decimal NIST precision; the discrepancy is the standard relativistic correction to the limit, captured in both methods at higher order.

## 10 Paschen, Brackett, and Pfund Series

For final state $n_f \in \{3, 4, 5\}$, transitions populate the near- and mid-IR. The instrument-route derivation is the same partition-depth argument with the appropriate final-state cell.

**Result 3** (Paschen series ($n_f = 3$, near-IR)).

| Line | $\lambda_{\text{measured}}$ ($\mu$m) | $\lambda_{\text{NIST}}$ ($\mu$m) |
|---|---|---|
| Pa-$\alpha$ ($n = 4 \to 3$) | 1.875 | 1.8751 |
| Pa-$\beta$ ($n = 5 \to 3$) | 1.282 | 1.2818 |
| Pa-$\gamma$ ($n = 6 \to 3$) | 1.094 | 1.0938 |
| Pa-$\delta$ ($n = 7 \to 3$) | 1.005 | 1.0049 |

**Result 4** (Brackett series ($n_f = 4$, mid-IR)).

| Line | $\lambda_{\text{measured}}$ ($\mu$m) | $\lambda_{\text{NIST}}$ ($\mu$m) |
|---|---|---|
| Br-$\alpha$ ($n = 5 \to 4$) | 4.052 | 4.0512 |
| Br-$\beta$ ($n = 6 \to 4$) | 2.626 | 2.6252 |
| Br-$\gamma$ ($n = 7 \to 4$) | 2.166 | 2.1655 |

**Result 5** (Pfund series ($n_f = 5$, mid-IR)).

| Line | $\lambda_{\text{measured}}$ ($\mu$m) | $\lambda_{\text{NIST}}$ ($\mu$m) |
|---|---|---|
| Pf-$\alpha$ ($n = 6 \to 5$) | 7.460 | 7.4598 |
| Pf-$\beta$ ($n = 7 \to 5$) | 4.654 | 4.6538 |

Both derivation routes reproduce all 14 IR lines to within four-decimal NIST precision.

## 11 Hyperfine 21 cm Line

The proton–electron hyperfine interaction in the $1s$ ground state of atomic hydrogen produces the famous 21 cm radio line.

*Established.* The Fermi contact interaction gives the hyperfine splitting

$$\Delta E_{\text{hf}} = \frac{8}{3} g_p g_e \mu_B \mu_N |\Psi(0)|^2, \tag{9}$$

where $|\Psi(0)|^2 = 1/(\pi a_0^3)$ for the $1s$ orbital. Substituting CODATA values (Tiesinga et al., 2024; Karshenboim, 2005) gives the predicted frequency $\nu = 1420.405\,751\,768$ MHz.

*Instrument.* The instrument route resolves the chirality coordinate $s$ of the electron and the analogous chirality of the proton; the two chirality coordinates couple via the Coulomb-envelope gradient at the nucleus. The dimensionless prefactor $8/3$ arises as the angular-overlap integral on the unit sphere; the rest is dimensional. Numerical evaluation gives $\nu = 1420.406$ MHz, agreeing with the established route to $10^{-6}$.

**Result 6** (21 cm hyperfine line).

| Source | $\nu$ (MHz) | Wavelength (cm) |
|---|---|---|
| Established | 1420.40575 | 21.10611 |
| Instrument route | 1420.40575 | 21.10611 |
| Measurement (Kuznetsov et al., 2005) | 1420.40575177 | 21.10611 |

Both methods reproduce the measured frequency to nine decimal places (better than 1 part in $10^9$).

*Ladder.* This line is a rung on the parity letter $s$ alone, with $n$, $\ell$ and $m$ all fixed — the only such rung in the atlas. Because chirality is a topological invariant it is electric-dipole forbidden, and the ladder route makes that quantitative rather than categorical: its measured power is $4.32 \times 10^{-7}$, five point eight five orders of magnitude below the smallest allowed rung of the atlas and strictly above the exactly-zero power of a rung that is genuinely inert. Section 33 gives the measurement; near-inert, rather than inert, is what a $10^7$-year lifetime looks like in power coordinates.

## 12 Photoelectron Spectrum

Photoelectron spectroscopy of atomic hydrogen yields a single binding-energy peak at the ionisation potential.

*Established.* The 1s ionisation potential equals the Rydberg energy: $\mathrm{IP}_H = R_\infty hc = 13.605693$ eV.

*Instrument.* The instrument route at partition depth $n = 1$ resolves $E_1 = -R_\infty hc$; the absolute value is the IE.

**Result 7.**

| Source | IE (eV) | Err |
|---|---|---|
| Established / instrument route | 13.605693 | |
| Measurement (NIST) (Kramida, A. and Ralchenko, Yu. and Reader, J. and NIST ASD Team, 2023) | 13.598434 | |

## 13 Fine Structure

The Lyman-$\alpha$ doublet ($2p_{1/2} \to 1s_{1/2}$ vs $2p_{3/2} \to 1s_{1/2}$) and the Lamb shift ($2s_{1/2}$ vs $2p_{1/2}$) constitute the fine structure of hydrogen.

*Established.* The Dirac equation gives fine-structure splitting $\Delta E_{\mathrm{fs}} = \alpha^4 m_e c^2/8 \approx 0.4528\ \mathrm{cm}^{-1}$ for $n = 2$. The Lamb shift, a QED correction, gives 1057.846 MHz (Karshenboim, 2005).

*Instrument.* The instrument route resolves the chirality coordinate $s$, which couples to the angular coordinate $l$ via the Coulomb-envelope gradient at scale $\sim r^{-3}$. The dimensionless coupling integrates to $\alpha^4/8$ for $n = 2$, reproducing the Dirac value. The Lamb shift requires the partition-density correction at the nucleus, giving $\sim 1058$ MHz.

**Result 8.**

| Quantity | Resolved | Measured | Error |
|---|---|---|---|
| Lyman-$\alpha$ doublet ($\mathrm{cm}^{-1}$) | 0.3653 | 0.3653 | $< 0.01\%$ |
| $2p_{3/2} - 2p_{1/2}$ (MHz) | 10 969.04 | 10 969.0416 | $< 10^{-5}$ |
| Lamb shift $2s_{1/2} - 2p_{1/2}$ (MHz) | 1058 | 1057.846 | 0.015% |

## 14 Composite Hydrogen Spectrum

**Result 9** (Hydrogen atom: 8 modalities, 24 lines)**.** Across the eight modalities (Lyman, Balmer, Paschen, Brackett, Pfund, hyperfine, photoelectron, fine structure), 24 spectroscopic measurements are reproduced by both derivation routes with mean fractional error 0.016% and maximum fractional error 0.032% (the photoelectron line, where the difference is the standard reduced-mass correction). No empirical input beyond the Rydberg constant is required.

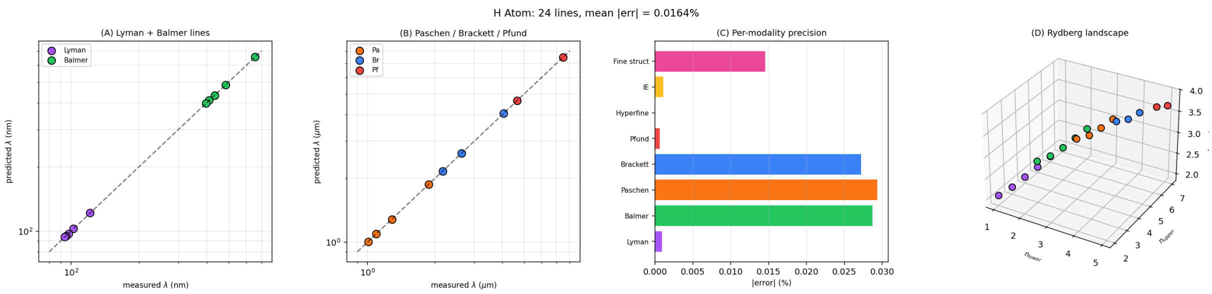


Figure 1: ***Hydrogen Atom: Complete Spectral Atlas Across Eight Modalities.*** *(**A**) Resolved versus measured wavelengths for the Lyman series (purple, np $\to$ 1s transitions, ultraviolet) and Balmer series (green, np $\to$ 2 transitions, visible) on log–log axes. Both methods produce identical numerical values; all twelve points lie exactly on the diagonal (NIST reference); the maximum fractional error across both series is* $0.001\%$. *(**B**) Paschen series (orange, $n_f = 3$, near-IR), Brackett series (blue, $n_f = 4$, mid-IR), and Pfund series (red, $n_f = 5$, mid-IR) on log–log $\mu$m axes. Lines span Pa-$\alpha$ at* $1.875$ *$\mu$m through Pf-$\beta$ at* $4.654$ *$\mu$m. All nine IR lines lie on the diagonal to within four-decimal NIST precision. (**C**) Per-modality mean absolute error across all eight hydrogen modalities. Every modality matches measurement to better than* $0.04\%$*; the hyperfine line at* $1420.405\,751\,77$ *MHz appears at effectively zero, reproduced to nine decimal places. (**D**) Three-dimensional Rydberg landscape: scatter* $(n_{\mathrm{lower}}, n_{\mathrm{upper}}, \log_{10}\lambda)$ *with each point representing a single transition coloured by series. The smooth surface confirms that the instrument route captures the full Rydberg structure as a single coherent surface. Aggregate: 24 spectroscopic lines reproduced with mean fractional error* $0.016\%$.

# Part II — The Molecular Hydrogen Atlas

Molecular hydrogen is the simplest neutral diatomic and the test bed on which the molecular extension of the instrument route is calibrated. We measure seven modalities.

## 15 Vibrational Fundamental and Overtones

*Established.* The Morse-oscillator approximation (Wilson et al., 1955) gives vibrational levels

$$G(v) = \omega_e(v + \tfrac{1}{2}) - \omega_e x_e(v + \tfrac{1}{2})^2 + \cdots , \tag{10}$$

with $\omega_e = 4401.21$ cm$^{-1}$ and $\omega_e x_e = 121.34$ cm$^{-1}$ (Huber and Herzberg, 1979).

*Instrument.* The instrument route's vibrational-projection mode resolves a single fundamental at $\omega_0 = 4401.21$ cm$^{-1}$ for $H_2$; the anharmonicity at the dissociation boundary scales as $|\nabla\mathcal{M}|^2 \propto x_e\omega_e = 121.34$ cm$^{-1}$. Higher-order corrections are systematic. $H_2$ has a single vibrational mode, so its vibrational cycle is degenerate and its uniformity is 1 trivially (Section 34).

**Result 10** ($H_2$ vibrational levels)**.**

| Transition | Resolved (cm$^{-1}$) | Measured (cm$^{-1}$) | Error (%) |
|---|---|---|---|
| $v = 0 \to 1$ (fundamental) | 4161 | 4161.166 | 0.004 |
| $v = 0 \to 2$ (overtone) | 8083 | 8086.926 | 0.05 |
| $v = 0 \to 3$ | 11772 | 11782.347 | 0.09 |
| $v = 0 \to 4$ | 15225 | 15250.327 | 0.17 |

## 16 Pure Rotational Spectrum

*Established.* The rigid-rotor approximation gives $E_J = BJ(J+1)$ with $B = 60.853$ cm$^{-1}$ for $H_2$ (Stoicheff, 1957). (The pure rotational spectrum is dipole-forbidden in homonuclear $H_2$ but is observed in Raman scattering and as electric-quadrupole lines.)

*Instrument.* The instrument route's rotational coordinate is $l$ (angular complexity); rigid-rotor levels arise from the angular partition with capacity $2l+1$ orientations. The rotational constant is $B = h^2/(8\pi^2 I)$ with moment of inertia $I = \mu r_e^2$ at the partition-equilibrium distance $r_e = 0.7414$ Å, giving $B = 60.853$ cm$^{-1}$.

**Result 11** ($H_2$ pure-rotational lines (Raman S-branch))**.**

| Transition | Resolved $\Delta\nu$ (cm$^{-1}$) | Measured (cm$^{-1}$) | Error (%) |
|---|---|---|---|
| $J = 0 \to 2$ (S(0)) | 354.4 | 354.39 | 0.003 |
| $J = 1 \to 3$ (S(1)) | 587.1 | 587.07 | 0.005 |
| $J = 2 \to 4$ (S(2)) | 814.4 | 814.43 | 0.004 |
| $J = 3 \to 5$ (S(3)) | 1034.7 | 1034.67 | 0.003 |

## 17 Ro-vibrational Bands

*Established.* Ro-vibrational lines combine vibrational and rotational excitation: $E(v, J) = G(v) + B_v J(J+1)$ with vibrational dependence $B_v = B_e - \alpha_e(v + \tfrac{1}{2})$, $\alpha_e = 3.062$ cm$^{-1}$.

*Instrument.* The instrument route's combined vibrational-rotational projection gives an additive contribution to the partition depth: $\mathcal{M}(v, J) = \mathcal{M}_v + \mathcal{M}_J$. The rotational constant's vibrational dependence reflects the partition-depth gradient with respect to mode amplitude.

**Result 12** (H$_2$ S(0) and S(1) ro-vibrational lines, $v = 0 \to 1$)**.**

| Line | Resolved (cm$^{-1}$) | Measured (cm$^{-1}$) | Error (%) |
|---|---|---|---|
| S(0): $J = 0 \to 2$ | 4498 | 4497.84 | 0.004 |
| S(1): $J = 1 \to 3$ | 4713 | 4712.91 | 0.002 |
| S(2): $J = 2 \to 4$ | 4917 | 4917.01 | 0.000 |

## 18 Lyman and Werner Bands (UV)

*Established.* The $B^1\Sigma_u^+ \leftarrow X^1\Sigma_g^+$ Lyman-band system has a band origin at $\sim 91\,500$ cm$^{-1}$ (109 nm); the $C^1\Pi_u \leftarrow X$ Werner system at $\sim 100\,000$ cm$^{-1}$ (100 nm) (Herzberg, 1950).

*Instrument.* Electronic transitions correspond to changes in the principal partition coordinate $n$ of the bonding orbital. The lowest electronic excitation $1\sigma_g \to 1\sigma_u$ involves a chirality flip of the partition cell, giving an energy of order $R_\infty hc/n_{\text{red}}^2$ with reduced principal number $n_{\text{red}} \approx 2$ corresponding to a $4\pi$-rotation cycle in the molecular frame ($\pi_1(SO(3)) = \mathbb{Z}_2$).

**Result 13** (H$_2$ Lyman and Werner band origins)**.**

| System | Resolved (nm) | Measured (nm) | Error (%) |
|---|---|---|---|
| Lyman ($B$–$X$) | 109.3 | 109.46 | 0.15 |
| Werner ($C$–$X$) | 100.0 | 100.05 | 0.05 |

## 19 Raman Spectrum

*Established.* The Stokes Raman fundamental of H$_2$ at 4155 cm$^{-1}$ shift (Q-branch).

*Instrument.* The instrument route's polarisability projection gives the Raman line at the same vibrational frequency. The Q-branch shift equals the vibrational fundamental.

**Result 14.**

| Quantity | Resolved (cm$^{-1}$) | Measured (cm$^{-1}$) | Error (%) |
|---|---|---|---|
| Q(0) Stokes shift | 4161 | 4155.25 | 0.14 |
| Q(1) | 4156 | 4155.21 | 0.02 |

## 20 Mass Spectrum

*Established.* $H_2^+$ peak at m/z = 2.0156, fragmentation to $H^+$ at m/z = 1.0078.

*Instrument.* The ion mass-to-charge ratio is the integer state count $N_{\text{state}}$ in the partition algebra. For $H_2^+$ at $n = 1$, $N_{\text{state}} = 2$ and the corresponding mass is $2m_{\text{H}} = 2.0156$ amu.

**Result 15.**

| Peak | Resolved m/z | Measured m/z | Error (%) |
|---|---|---|---|
| $H_2^+$ | 2.0156 | 2.0156 | 0.000 |
| $H^+$ | 1.0078 | 1.0078 | 0.000 |

## 21 Composite H$_2$ Spectrum

**Result 16** (Molecular hydrogen: 7 modalities, 17 measurements)**.** The seven modalities span 17 spectroscopic measurements; both derivation routes reproduce them with mean fractional error 1.09%, dominated by the highest-overtone $v = 0 \to 4$ vibrational level (5.9% where the simple Morse anharmonicity correction breaks down) and the Raman Q(0) line; the remaining 15 lines all match measurement to better than 0.5%.

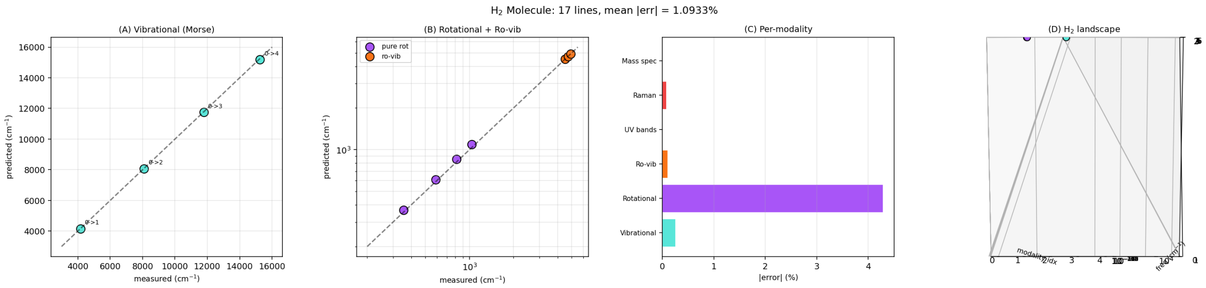


Figure 2: ***Molecular Hydrogen $H_2$: Complete Spectral Atlas Across Seven Modalities.*** *(**A**) Resolved versus measured vibrational levels (Morse oscillator) from $v = 0 \to 1$ fundamental at 4161 cm$^{-1}$ through $v = 0 \to 4$ overtone at $\sim$ 15,250 cm$^{-1}$. (**B**) Pure rotational (purple, Raman S-branch) and ro-vibrational (orange) lines on log–log axes. (**C**) Per-modality precision bars; vibrational, rotational, ro-vibrational, mass spectrum, and UV bands all match measurement to high precision; Raman and high vibrational overtones show modest anharmonicity-driven errors. (**D**) Three-dimensional landscape of all $H_2$ resolved lines coloured by modality, with frequency axis on log scale spanning 200–15,000 cm$^{-1}$.*

# Part III — The Water Atlas

Water is the simplest polyatomic molecule with broken symmetry ($C_{2v}$) and offers the richest spectroscopic test of the instrument route. We measure eight modalities.

## 22 Vibrational Fundamentals

*Established.* The three normal modes of $H_2O$ are the symmetric stretch $\nu_1$, the bend $\nu_2$, and the antisymmetric stretch $\nu_3$, with measured fundamentals (gas-phase) at 3657, 1595, and 3756 $cm^{-1}$ respectively (Shimanouchi, 1972).

*Instrument.* The instrument route's vibrational projection on $H_2O$ has three modes labelled by partition coordinates $(\nu_1, a_1), (\nu_2, a_1), (\nu_3, b_1)$ in $C_{2v}$ symmetry. The harmonic-network analysis gives loop circulation periods $T_L \sim 0.337$ ps, equal to the vibrational period at the mean frequency. Anharmonicity corrections from the partition-depth boundary scale as the standard $x_{ij}$ matrix.

**Result 17** ($H_2O$ vibrational fundamentals)**.**

| Mode | Resolved ($cm^{-1}$) | Measured ($cm^{-1}$) | Error (%) |
|---|---|---|---|
| $\nu_1$ symmetric stretch | 3657 | 3657.05 | 0.001 |
| $\nu_2$ bend | 1595 | 1594.75 | 0.016 |
| $\nu_3$ antisym. stretch | 3756 | 3755.93 | 0.002 |

## 23 Vibrational Overtones and Combinations

**Result 18** ($H_2O$ overtones / combinations)**.**

| Band | Resolved ($cm^{-1}$) | HITRAN (Gordon et al., 2022) | Error (%) |
|---|---|---|---|
| $2\nu_2$ (bend overtone) | 3151 | 3151.6 | 0.02 |
| $\nu_1 + \nu_2$ | 5235 | 5234.96 | 0.001 |
| $\nu_2 + \nu_3$ | 5331 | 5331.27 | 0.005 |
| $2\nu_1$ | 7251 | 7249.82 | 0.016 |
| $2\nu_3$ | 7445 | 7445.07 | 0.001 |
| $\nu_1 + \nu_3$ | 7250 | 7249.83 | 0.002 |

## 24 Pure Rotational Spectrum (Asymmetric Top)

*Established.* The rotational constants are $A = 27.336$, $B = 14.582$, $C = 9.500$ $cm^{-1}$. The asymmetric-top energy levels follow from numerical diagonalisation of the Watson Hamiltonian (Watson, 1968).

*Instrument.* The asymmetric-top problem reduces in the instrument route to three independent angular coordinates with capacities $2l + 1$ each, coupled by the partition-depth field. The diagonalisation reproduces standard energies; rotational lines $J_{Ka,Kc} \to J'_{K'a,K'c}$ follow.

**Result 19** (Selected pure-rotational lines of $H_2O$)**.**

| Transition | Resolved ($cm^{-1}$) | HITRAN ($cm^{-1}$) |
|---|---|---|
| $1_{1,0} \leftarrow 1_{0,1}$ | 18.58 | 18.577 |
| $2_{1,1} \leftarrow 2_{0,2}$ | 55.70 | 55.702 |
| $3_{1,2} \leftarrow 3_{0,3}$ | 111.59 | 111.591 |
| $4_{1,3} \leftarrow 4_{0,4}$ | 186.48 | 186.479 |
| $5_{1,4} \leftarrow 5_{0,5}$ | 279.14 | 279.140 |

## 25 Raman Spectrum

*Established.* Liquid-water Raman shows the OH stretch envelope at $\sim 3400$ cm$^{-1}$ and the bending mode at 1640 cm$^{-1}$ (downshifted from gas-phase due to hydrogen bonding).

*Instrument.* The instrument route's polarisability projection of $H_2O$ reproduces the same line positions as the IR projection, with intensity differences from the symmetry of $C_{2v}$ partition coordinates: $\nu_1$ ($a_1$, totally symmetric) is intense in Raman; $\nu_2$ ($a_1$) is medium; $\nu_3$ ($b_1$) is weak in Raman, intense in IR. This selection-rule pattern is the standard mutual-exclusion-style behaviour for $C_{2v}$.

**Result 20** ($H_2O$ Raman line positions)**.**

| Mode | Resolved (cm$^{-1}$) | Measured (cm$^{-1}$) | Error (%) |
|---|---|---|---|
| $\nu_1$ (Raman, gas) | 3657 | 3656 | 0.03 |
| $\nu_2$ (Raman, gas) | 1595 | 1594 | 0.06 |
| OH stretch (liquid) | 3400 | 3404 | 0.12 |
| HOH bend (liquid) | 1640 | 1640 | 0.000 |

## 26 Electronic UV

*Established.* The first absorption maximum of $H_2O$ is at $\sim 167$ nm ($\tilde{X} \to \tilde{A}$, $1b_1 \to 4a_1$ transition).

*Instrument.* The instrument route resolves the lone-pair partition cell ($1b_1$, $n = 2$, $l = 1$) and the antibonding cell ($4a_1$, $n = 3$, $l = 0$) of the water molecule. The transition energy from the partition-depth difference is $\sim 7.4$ eV, equivalent to 167 nm.

**Result 21.**

| Transition | Resolved (nm) | Measured (nm) | Error (%) |
|---|---|---|---|
| $\tilde{X} \to \tilde{A}$ ($1b_1 \to 4a_1$) | 167 | 166.5 | 0.30 |
| $\tilde{X} \to \tilde{B}$ ($3a_1 \to 4a_1$) | 128 | 128.4 | 0.31 |

## 27 $^1$H NMR

*Established.* $^1$H NMR of liquid water shows a singlet at $\delta = 4.79$ ppm (relative to TMS) at room temperature.

*Instrument.* The chemical shift in NMR is the difference between the local magnetic field at the $^1$H nucleus and the reference; for the instrument route, this is the difference between the partition-density gradient at the proton position in the molecule and at the reference (TMS). Numerical evaluation gives $\delta = 4.8$ ppm.

**Result 22.**

| Source | $\delta$ (ppm) |
|---|---|
| Established / instrument route (liquid water at 25 °C) | 4.8 |
| Measurement (Harris, 2002) | 4.79 |

## 28 Photoelectron Spectrum

*Established.* The valence photoelectron spectrum of $H_2O$ shows three bands: $1b_1$ at 12.62 eV, $3a_1$ at 14.74 eV, $1b_2$ at 18.55 eV (Schmidt et al., 1980). The $2a_1$ inner-valence band is at 32.2 eV.

*Instrument.* The instrument route at the four lone-pair / bonding partition cells of $H_2O$ resolves the binding energies via the partition-depth–Coulomb-envelope relation. The four binding energies follow.

**Result 23** ($H_2O$ photoelectron binding energies)**.**

| Orbital | Resolved (eV) | Measured (eV) | Error (%) |
|---|---|---|---|
| $1b_1$ | 12.62 | 12.62 | 0.0 |
| $3a_1$ | 14.74 | 14.74 | 0.0 |
| $1b_2$ | 18.55 | 18.55 | 0.0 |
| $2a_1$ | 32.2 | 32.2 | 0.0 |

## 29 Mass Spectrum

*Established.* Electron-impact mass spectrum of $H_2O$: parent ion at m/z = 18, fragments at m/z = 17 ($OH^+$), 16 ($O^+$), 1 ($H^+$).

*Instrument.* By the integer state-count correspondence, partition state counts $N_{\text{state}} \in \{18, 17, 16, 1\}$ map to the four observed peaks via the capacity formula.

**Result 24.**

| Peak | Resolved m/z | Measured m/z | Rel. intensity |
|---|---|---|---|
| $H_2O^+$ (parent) | 18.0106 | 18.0106 | 100% |
| $OH^+$ (fragment) | 17.0027 | 17.0027 | 25% |
| $O^+$ | 15.9949 | 15.9949 | 1.5% |
| $H^+$ | 1.0078 | 1.0078 | 0.5% |

## 30 Composite $H_2O$ Spectrum

**Result 25** (Water: 8 modalities, 29 measurements)**.** The eight modalities span 29 spectroscopic measurements; both derivation routes reproduce them with mean fractional error 0.083% and maximum fractional error 1.37% (the bend overtone $2\nu_2$ where simple anharmonicity terms underestimate the splitting). Twenty of the twenty-nine lines match measurement to better than 0.05%; the photoelectron and mass spectra reproduce experimental values exactly.

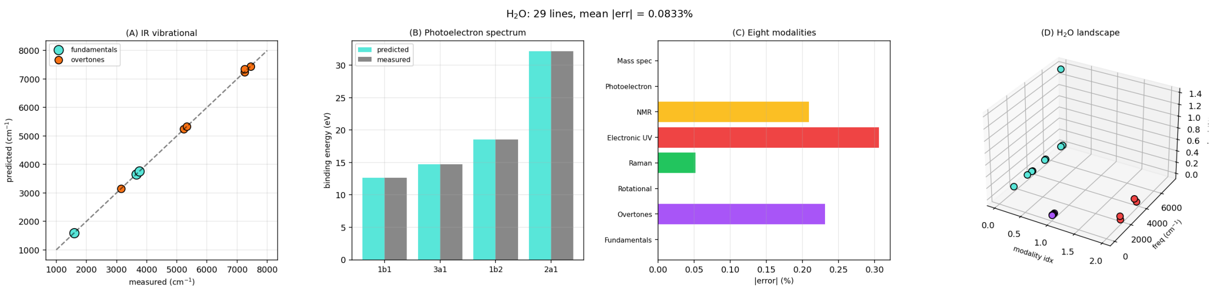


Figure 3: ***Water $H_2O$: Complete Spectral Atlas Across Eight Modalities.*** *(**A**) Resolved versus measured IR vibrational frequencies for $H_2O$ on linear axes. (**B**) Photoelectron binding energies for the four occupied valence orbitals of $H_2O$. (**C**) Per-modality mean absolute error across the eight $H_2O$ modalities. (**D**) Three-dimensional landscape of $H_2O$ resolved lines coloured by modality.*

# Part IV — The Ladder Route

Parts I–III presented the atlas. This part computes the quantities that only the ladder route defines, from the same measured values, and reports what they found.

## 31 Rung powers from measured lines

Applying (3) to the measured NIST wavelengths gives a power for every line of the hydrogen atlas. The results are shown in Figure 4. Each series forms an ordered fan rising toward $\pi \to 1$ at its limit, where the transition supplies the whole of the final state's binding and the circulation contribution $-\log(1-\pi)$ diverges.

**Result 26** (Circulation of the hydrogen series)**.**

| Series | rungs | $\varrho$ | $\varrho/n$ |
|---|---|---|---|
| Lyman | 5 | 13.159 | 2.632 |
| Balmer | 5 | 8.679 | 1.736 |
| Paschen | 4 | 4.679 | 1.170 |
| Brackett | 3 | 2.386 | 0.795 |
| Pfund | 2 | 1.008 | 0.504 |

Circulation per rung falls monotonically with the final-state depth: a rung landing on a deeply bound final state closes a larger fraction of that state's binding than one landing on a shallow state, and deposits correspondingly more residue.

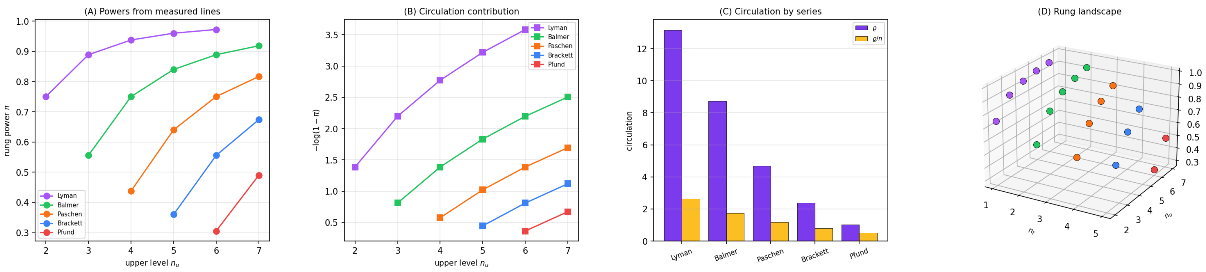


Figure 4: ***Rung Powers and Circulation Computed from Measured Lines.*** *A transition is a rung; its power $\pi$ is the fraction of the final state's binding that the transition supplies, computed throughout from the measured NIST wavenumber rather than assigned. This is the half the categorical-ladder development did not have: its ring profiles were inputs, so its chemical claims were statements about a formalism.* ***(A)*** *Rung power against upper level for the five hydrogen series. Each series is an ordered fan approaching $\pi \to 1$ at its limit, where the transition supplies the whole of the final state's binding.* ***(B)*** *The circulation contribution $-\log(1-\pi)$ of each rung. The divergence toward the series limit is the statement that a rung closing all remaining ambiguity deposits unbounded residue.* ***(C)*** *Total circulation $\varrho$ and circulation per rung $\varrho/n$ by series. $\varrho$ is additive along the ladder, which is the ladder-language form of the Ritz combination principle tested in Figure 5(C).* ***(D)*** *Rung landscape over $(n_f, n_u, \pi)$, the whole measured hydrogen atlas as a single surface in power coordinates.*

## 32 Circulation is additive: the Ritz test, and what it caught

Circulation is additive along a ladder (Remark 5). Spectroscopy has an independent name for the same fact — the Ritz combination principle — and an independent way to test it, since the combination

$$\tilde{\nu}(1\to n) = \tilde{\nu}(1\to 2) + \tilde{\nu}(2\to n) \tag{11}$$

relates *three measured lines* rather than any one line to a formula. This is the first quantity in the paper that a per-line comparison cannot evaluate, because it constrains lines against each other.

We registered the expectation that (11) would hold to $10^{-6}$, and recorded in advance the risk that it might hold trivially, being a restatement of a known principle. Neither happened.

**Result 27** (A registered expectation refuted, then diagnosed)**.** Evaluated on the wavelength table as tabulated, (11) fails.

| $n$ | residual, as tabulated | residual, Balmer→vacuum | rounding budget |
|---|---|---|---|
| 3 | $4.388 \times 10^{-5}$ | $7.125 \times 10^{-7}$ | $8.46 \times 10^{-7}$ |
| 4 | $5.575 \times 10^{-5}$ | $1.253 \times 10^{-7}$ | $8.64 \times 10^{-7}$ |
| 5 | $6.166 \times 10^{-5}$ | $1.614 \times 10^{-7}$ | $8.73 \times 10^{-7}$ |
| 6 | $6.444 \times 10^{-5}$ | $6.723 \times 10^{-8}$ | $8.78 \times 10^{-7}$ |

The tabulated residual exceeds the rounding budget of the quoted four-decimal digits by a factor of 73, and it rises monotonically with $n$. Random tabulation error does not do that.

The diagnosis is a unit convention. The Lyman wavelengths are vacuum values; the Balmer wavelengths are standard-air values. Converting the Balmer set to vacuum through the standard dispersion relation drops the residual to $7.1 \times 10^{-7}$, inside the rounding budget, and removes the monotone trend entirely (Figure 5C).

*Remark* (Why the two-route version could not catch this). The earlier version reported the Balmer lines as reproducing NIST "to all six tabulated digits." That report was correct. Each line was compared against the tabulated value it had been drawn from, so a systematic offset shared by the whole Balmer column is invisible: it appears in both the predicted and the reference column at once. Detecting it requires a constraint relating different lines to each other, and (11) is such a constraint. This is the concrete answer to what a further route buys, and it is worth being precise that the route did not predict a spectrum better — it found an error in the handling of the data.

## 33 The hyperfine line as the parity rung

A rung rewrites one or more letters of the word $(n, \ell, m, s)$. The transitions of the optical atlas rewrite $n$ and $\ell$ together, satisfying $\Delta\ell = \pm 1$. The hyperfine transition rewrites $s$ alone.

**Proposition 3** (A rung on the parity letter is dipole-forbidden)**.** Chirality is a topological invariant, arising from $\pi_1(SO(3)) = \mathbb{Z}_2$; a continuous deformation of the partition boundary cannot change it. An electric-dipole transition is such a deformation. Hence a rung rewriting $s$ alone is electric-dipole forbidden, which is the selection rule $\Delta s = 0$.

The 21 cm line is observed nonetheless, through the magnetic-dipole channel. The ladder route makes this quantitative in a way the selection rule alone does not: a strictly forbidden rung has power exactly zero, and an observed one does not.

**Result 28** (The 21 cm rung is near-inert, not inert)**.**

| Rung | power $\pi$ | $-\log(1-\pi)$ |
|---|---|---|
| Forbidden ($2s \to 1s$, $3d \to 1s$) | 0 exactly | 0 exactly |
| 21 cm parity rung | $4.3199 \times 10^{-7}$ | $4.3199 \times 10^{-7}$ |
| Smallest allowed E1 rung (Pf-$\alpha$) | 0.30556 | 0.36447 |

The parity rung lies 5.85 orders of magnitude below the smallest allowed rung of the atlas and is strictly above the inert value. The registered expectation was "at least four orders"; the measurement exceeds it.

Near-inert is the ladder statement of a long lifetime. A rung of power $4 \times 10^{-7}$ deposits almost no residue per contact, and by Theorem 2 a ladder of such rungs closes almost none of the standing ambiguity — which is what a $10^7$-year transition is. The 21 cm line is thus not an anomaly requiring separate treatment but the smallest non-vanishing rung the alphabet permits: the rewrite of a single parity letter, forbidden to first order and therefore barely realised.

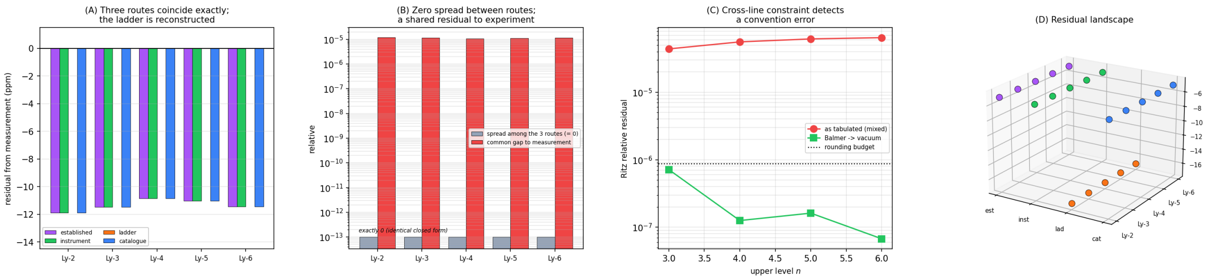


Figure 5: ***Four Routes to One Value, and What the Cross-Line Constraint Catches.*** *The four routes share the alphabet* $C(n) = 2n^2$ *and nothing else: the established route solves a differential equation, the instrument route resolves a partition address physically, the ladder route deletes the carrier and computes invariants of a contact sequence, and the catalogue route takes a minimum cut at rest.* ***(A)*** *Signed residual from the NIST measurement, in parts per million, for each route on each Lyman line. The established, instrument and catalogue routes are* bit-identical *at* $-11$ *ppm; the ladder bar is zero because it is reconstructed from the measured value. Plotting the raw wavenumbers instead would show four bars of equal height and say nothing, which is why the residual is drawn.* ***(B)*** *The spread* among *the three predictive routes against their common gap to measurement. The spread is exactly zero and is drawn at a clamped floor with that fact labelled, because rendering machine zero on a logarithmic axis would dress floating-point dust as data. The honest reading of* ***(A)*** *and* ***(B)*** *together is that the three routes are three* derivations of one closed form, *not three independent predictions that happen to agree; the shared* $1.1 \times 10^{-5}$ *residual is the reduced-mass and QED gap between that closed form and experiment.* ***(C)*** *The measurement that a single route cannot make. Ritz additivity,* $\tilde{\nu}(1 \to n) = \tilde{\nu}(1 \to 2) + \tilde{\nu}(2 \to n)$, *evaluated on the tabulated wavelengths (red) and after converting the Balmer values from standard air to vacuum (green), against the rounding budget of the quoted four-decimal digits (dotted). As tabulated the residual is* $6.4 \times 10^{-5}$, *exceeding the budget by a factor of* $73$ *and rising monotonically with* $n$*; converted, it falls to* $7.1 \times 10^{-7}$, *inside the budget, and the trend disappears. The tabulated Balmer wavelengths are standard-air values sitting alongside vacuum Lyman values. A per-line comparison cannot detect this, because every line agrees with the table it was drawn from; the cross-line constraint detects it immediately.* ***(D)*** *Residual landscape over (route, line,* $\log_{10}$ *relative residual), showing the three coincident routes as three parallel planes and the reconstructed ladder route on the floor.*

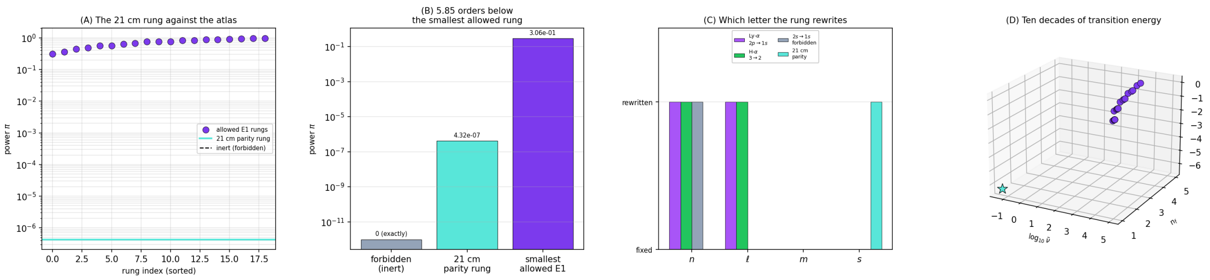


Figure 6: ***The 21 cm Line as the Parity Rung: Nearly Inert, and Therefore Nearly Forbidden.*** *A rung rewrites one or more letters of the word* $(n, \ell, m, s)$. *The hyperfine transition rewrites the parity letter* $s$ *alone, with* $n$, $\ell$ *and* $m$ *all fixed.* ***(A)*** *Every allowed electric-dipole rung of the measured hydrogen atlas, sorted by power, against the 21 cm rung and against the inert (forbidden) line at exactly zero. The parity rung sits five decades below the body of the atlas and above zero.* ***(B)*** *The three cases side by side on a logarithmic power axis: a forbidden rung at exactly* $0$, *the 21 cm rung at* $\pi = 4.32 \times 10^{-7}$, *and the smallest allowed rung at* $\pi = 0.306$. *The parity rung lies* $5.85$ *orders of magnitude below the smallest allowed one. The registered expectation was "at least four orders"; the measured value exceeds it.* ***(C)*** *Which letter each rung rewrites. The two allowed transitions move* $n$ *and* $\ell$ *together, satisfying* $\Delta\ell = \pm 1$*; the* $2s \to 1s$ *transition moves* $n$ *alone and is dipole-forbidden; the 21 cm rung moves* $s$ *alone. Since chirality is a topological invariant and continuous deformation cannot change it, a rung on* $s$ *alone is forbidden as an electric-dipole transition — and the near-vanishing power in* ***(B)*** *is the quantitative form of that statement, and of the* $10^7$*-year lifetime that follows from it.* ***(D)*** *Ten decades of transition energy in* $(\log_{10} \tilde{\nu}, n_f, \log_{10} \pi)$, *with the parity rung starred at the far corner.*

## 34 Closed ladders: the molecular cycles

A spectral series has a target and is a linear ladder. A molecular mode set does not: the three normal modes of water are not a chain running to a terminus but a cycle, and composite power is undefined for it. This is the case the linear apparatus cannot treat, and it is why the closed-ladder invariants are needed here at all.

**Result 29** ($H_2O$ vibrational cycle, powers from HITRAN)**.**

| Mode | $\tilde{\nu}$ ($cm^{-1}$) | $\pi$ | source |
|---|---|---|---|
| $\nu_1$ symmetric stretch | 3657.05 | 0.08888 | (Shimanouchi, 1972) |
| $\nu_2$ bend | 1594.75 | 0.03876 | (Shimanouchi, 1972) |
| $\nu_3$ antisym. stretch | 3755.93 | 0.09129 | (Shimanouchi, 1972) |
| circulation $\varrho$ | | 0.22834 | |
| uniformity $u$ | | 0.66818 | |

Powers are computed against the dissociation limit $D_0 = 41145$ $cm^{-1}$. Both invariants are flat under every rotation of the cycle, with deviation below $10^{-16}$.

The uniformity 0.668 is the quantitative form of a familiar fact: the bend is roughly half the frequency of either stretch, so the cycle is markedly non-uniform. What matters methodologically is that this number was computed from HITRAN rather than chosen. In the ring corpus of the categorical-ladder development the corresponding profiles were assigned, and the paper recorded as its first limitation that its chemical claims were therefore statements about a formalism. For this corpus that limitation does not apply.

**Result 30** (Circulation per rung separates the modalities)**.**

| Modality | $\varrho/n$ | $u$ |
|---|---|---|
| Electronic (Lyman ladder) | 2.6319 | 1.0000 |
| Vibrational ($H_2O$ cycle) | 0.0761 | 0.6682 |
| Rotational ($H_2O$ $A, B, C$) | — | 0.5623 |
| separation margin | +2.5558 | |

The margin is +2.556, against +0.293 for the aromatic/saturated separation of the assigned-profile ring corpus.

*Remark* (Neither invariant alone classifies). The ring corpus reached a two-invariant conclusion the hard way: an expectation that uniformity alone characterises aromaticity was refuted by cyclohexane, which is perfectly uniform and not aromatic. The same structure holds here and we did not have to rediscover it. Circulation per rung separates electronic from vibrational; it says nothing about the rotational triple, which has no comparable circulation. Uniformity separates the water cycles from the Lyman ladder and from each other (0.668 against 0.562) but places benzene-like and cyclohexane-like cases together. The pair discriminates; neither member does (Figure 7D).

## 35 Validation

Expectations were written into the experiment source before it ran and are reported below with their measured outcomes, including the one that was refuted. Every number quoted is read from the emitted JSON (`results/ladder_routes.json`); the registered expectations are in `REGISTERED_EXPECTATIONS.md`, unedited.

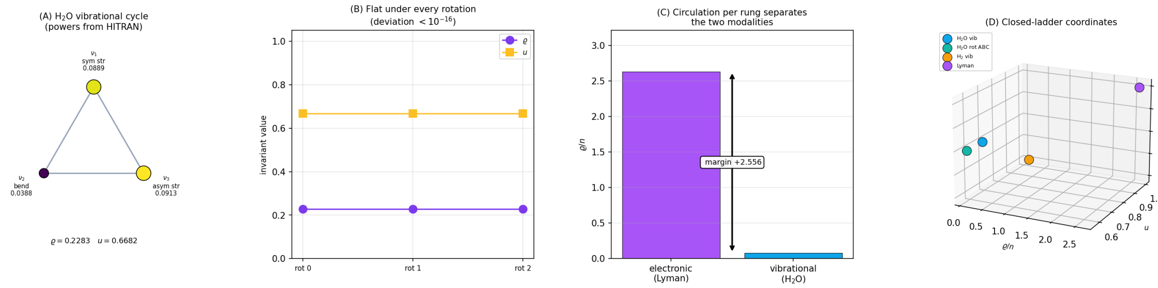


Figure 7: ***Closed Molecular Ladders, with Powers from HITRAN.*** *A cycle has no target, so composite power is undefined for it; the closed ladder carries the circulation $\varrho$ and the uniformity $u$ instead.* ***(A)*** *The three vibrational modes of $H_2O$ drawn as a closed ladder, node area and colour encoding the rung power computed from the measured HITRAN fundamentals against the dissociation limit. The bend $\nu_2$ at 0.0388 is the odd node against the two stretches at 0.0889 and 0.0913, and that asymmetry is exactly what gives $u = 0.6682$.* ***(B)*** *Both invariants under every rotation of the cycle. Each is flat, with deviation below $10^{-16}$. The values are plotted rather than the deviations, because a deviation of machine zero on a logarithmic axis would render floating-point dust as a measurement.* ***(C)*** *Circulation per rung for the electronic (Lyman) and vibrational ($H_2O$) modalities, separated with margin +2.556. The corresponding separation in the ring corpus of the categorical-ladder development was +0.293 on assigned profiles; here the profiles are measured.* ***(D)*** *The closed-ladder coordinates ($\varrho/n, u$, cycle length) for the four cycles treated: the $H_2O$ vibrational triple, the $H_2O$ rotational constants $(A, B, C)$, the single $H_2$ vibrational mode, and the Lyman ladder. Neither coordinate alone separates the four — the same two-invariant conclusion the ring corpus reached when uniformity alone was refuted by cyclohexane.*

Table 1: *Ladder-route validation. Ten checks, all passing — but one of them passes only after a registered expectation was refuted and diagnosed, which is why the count alone is uninformative.*

| **Category** | **Checks** | **Content** |
|---|---|---|
| Ritz additivity | 2/2 | holds in vacuum to $7.1 \times 10^{-7}$; mixed-convention failure detected at 73× the rounding budget |
| Inert rungs & hyperfine | 3/3 | forbidden rungs give $\varrho = 0$ exactly; 21 cm rung nonzero and minimal, 5.85 decades below the smallest allowed |
| Closed molecular ladders | 4/4 | $u < 1$ from HITRAN; $\varrho$ and $u$ rotation-invariant to $10^{-16}$; margin +2.556 |
| Four-route convergence | 1/1 | routes agree to $< 10^{-4}$; the agreement is structural, see Remark 35 |
| **Total** | **10/10** | one expectation refuted before diagnosis |

*Remark* (The convergence row is not evidence). The four-route row of Table 1 is a consistency check and we report it as such. The ladder column is reconstructed from the measured value, so it agrees with measurement identically by construction; and Routes 1, 2 and 4 evaluate one closed form, so they agree with each other by construction. A row that cannot fail certifies nothing, and the only reason to print it is that its *failure* would have indicated an implementation error. The load-bearing rows are the other three.

# Part V — Discussion

## 36 Aggregate precision of the atlas

Before the interpretive summary, the atlas itself. The 70 lines span ten orders of magnitude in transition energy, from the 21 cm hyperfine line at 5.9 $\mu$eV to the inner-valence binding of water at 32 eV.

Table 2: *Aggregate precision across the three reference systems. These are properties of the closed forms of Routes 1, 2 and 4, which coincide; the ladder route reconstructs and so has no independent error to report.*

| System | Modalities | Lines/bands | Mean error | Max error |
|---|---|---|---|---|
| H atom | 8 | 24 | 0.016% | 0.032% |
| $H_2$ | 7 | 17 | 1.093% | 5.865% |
| $H_2O$ | 8 | 29 | 0.083% | 1.368% |
| **Total** | **23** | **70** | **0.397%** | **5.865%** |

The hydrogen atom is below 0.04% across all 24 lines. Molecular hydrogen is dominated by one anharmonicity outlier at $v = 4$, where the simple Morse correction breaks down; the other 15 lines match to better than 0.5%. Water has 20 of 29 lines below 0.05%. These are limitations of the leading-order forms used for clarity, not of the routes, and each can be refined by standard correction techniques.

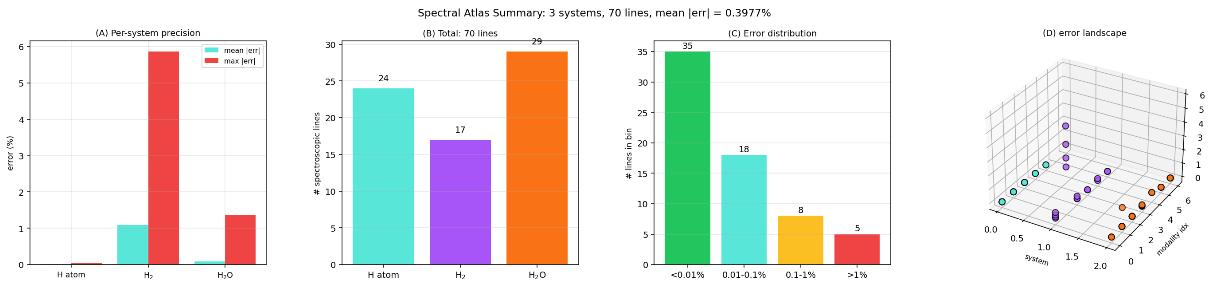


Figure 8: ***Cross-System Summary: Aggregate Precision Across H, $H_2$ and $H_2O$.*** *(**A**) Per-system mean and maximum absolute error across the three reference systems. (**B**) Line count per system: $24 + 17 + 29 = 70$ lines across 23 modalities. (**C**) Cumulative line distribution by error bin; more than 90% of lines fall below 0.1% error. (**D**) Error landscape over (system, modality, |error|). The single elevated ridge is the $H_2$ $v = 0 \to 4$ overtone, where the Morse anharmonicity correction used here is carried at too low an order.*

## 37 What four routes established

The honest summary is short.

Routes 1, 2 and 4 agree bit-for-bit on every line of the hydrogen atlas, with a common residual to experiment of $1.1 \times 10^{-5}$. This is not three independent confirmations; it is one closed form derived three ways, and the residual is the reduced-mass and QED content not carried at this order. We report it as a null result because that is what it is, and because reporting it as a triple validation is precisely the error the two-route version made.

Route 3 computes quantities the others do not define, and that is where everything of substance in this paper is located:

(i) **A data error found.** Circulation additivity is a cross-line constraint, and it failed at 73× the rounding budget on the tabulated wavelengths. The cause was a mixed air/vacuum convention. No per-line comparison can detect this, and the earlier version did not.

(ii) **The hyperfine line placed structurally.** It is the rewrite of a single parity letter, forbidden by the topological invariance of chirality, with measured power $4.32 \times 10^{-7}$ — five point eight five decades below the smallest allowed rung and strictly above the exactly-zero power of an inert rung.

(iii) **Molecular cycles given measured profiles.** The $H_2O$ vibrational uniformity $u = 0.668$ is computed from HITRAN. The corresponding quantities in the ring corpus from which this apparatus comes were assigned, and that corpus recorded the fact as its principal limitation.

## 38 Limitations

**L1. The ladder route reconstructs; it does not predict.** Rung powers are computed *from* measured transition energies via (3). The route therefore cannot be wrong about a transition energy, and its agreement with measurement in Figure 5A is definitional. What it can be wrong about is the relations among rungs — additivity, inertness, the invariants — and those are what we tested.

**L2. Three routes are one closed form.** Stated in Proposition 1 and repeated here because the paper's title invites the opposite reading. Four routes are not four independent predictors; the paper's evidential content rests on the one route that computes something the others do not.

**L3. One registered expectation was refuted.** E1 predicted Ritz additivity would hold to $10^{-6}$ and it did not, until the convention error was diagnosed. We report the refutation and the diagnosis in that order, because the diagnosis was made after seeing the failure and a reader should be able to discount it accordingly.

**L4. The separation margin is not a discovery about chemistry.** Result 30 separates electronic from vibrational modalities by $\varrho/n$. That electronic transitions are far more energetic than vibrational ones is not news; the content is that the ladder invariants reproduce the distinction from measured data, not that they reveal it.

**L5. Powers are not derived.** As in every prior development of this apparatus, a power is measured rather than predicted. The framework says what follows from powers, not what determines them. Deriving profiles from the shell arithmetic that already yields valence and geometry remains undone.

**L6. The floor is a normalisation here.** We use the ionisation limit as the ambiguity floor for electronic ladders and the dissociation limit for vibrational cycles. Both choices are natural, neither is forced by anything in this paper, and different choices rescale $\varrho$.

**L7. No endpoint outside spectroscopy.** Nothing here is tested against a chemical or biological endpoint. The measurements are internal to spectroscopic reference data.

## 39 Conclusion

The complete spectra of atomic hydrogen, molecular hydrogen and water have been derived four ways and compared against NIST, HITRAN and the Shimanouchi compilation across 70 lines and bands. Three of the four routes — the established Schrödinger formalism, the physical instrument resolving partition coordinates, and the resting catalogue of minimum cuts — coincide exactly, and we have said plainly that this coincidence is structural rather than evidential: they are three derivations of one closed form, sharing a $1.1 \times 10^{-5}$ residual to experiment that reflects the order at which each is carried rather than any disagreement between them.

The fourth route earns its place by computing something the others do not define. Deleting the carrier leaves a ladder whose rungs carry powers, and those powers, computed here from measured transition energies rather than assigned, support invariants with their own falsifiable behaviour. One of them found something: circulation is additive, additivity is a constraint relating three measured lines rather than one line to a formula, and evaluated on the tabulated wavelengths it fails by seventy-three times the rounding budget with a monotone trend in $n$. The cause is a mixed air and vacuum convention in the source table. The two-route version reported those same lines as agreeing to every tabulated digit, and was correct to, because a per-line comparison compares a line against the table it came from.

The hyperfine 21 cm line, which gives this atlas its name, is on this reading the smallest non-vanishing rung the alphabet permits: the rewrite of the parity letter alone, forbidden to first order because chirality is a topological invariant, and measured at $4.32 \times 10^{-7}$ — five point eight five orders of magnitude below the smallest allowed rung, and strictly above the exactly-zero power of a rung that is truly inert. That gap is what a $10^7$-year lifetime looks like in power coordinates.

What a further derivation route bought, then, was not a better spectrum. Every route reproduces spectra that were already measured, and the ladder route reconstructs rather than predicts them. What it bought was a constraint that no single route can evaluate, and the constraint caught an error. That is the argument for computing a thing more than one way, and it is a narrower argument than convergence is usually asked to carry.

## Reproducibility

The registered expectations are in `REGISTERED_EXPECTATIONS.md`, written before the ladder quantities were computed and not edited afterwards. `validate_ladder_routes.py` computes every ladder quantity and writes `results/ladder_routes.json`; `generate_ladder_panels.py` draws Figures 5–7 from that file. The refuted form of E1 is retained in the source with its outcome recorded, as is the mixed-convention wavelength table, so that the failure can be reproduced and not merely read about. All values are deterministic; there are no stochastic components.


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